Dhiraj Saw

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Date of Birth: 07.07.2002

EDUCATION

Doctor of Philosophy (Mathematics) 2024 – Present

Indian Institute of Technology, Goa, India

- ✓ *Advisor*: Dr. Lok Pati Tripathi
- ✓ *Current CPI*: 8.61/10 (2nd Year)
- ✓ *Related coursework*: Machine Learning, Advanced Partial Differential Equations, Stochastic Calculus and Applications, Stochastic PDEs - Theory & Numerics, Numerical Approximations of PDEs, Finite Element Methods.

Master of Science 2024

Indian Institute of Technology, Bhubaneswar(Odisha), India

- ✓ *CGPA*: 8.36/10
- ✓ *Related coursework*: Mathematics

Bachelor of Science 2022

Dr. Shyama Prasad Mukherjee University, Ranchi(Jharkhand), India

- ✓ *CGPA*: 8.94/10
- ✓ *Related coursework*: Major in Mathematics with minor courses in Physics

Intermediate 2019

St. Xavier's College, Ranchi(Jharkhand), India

- ✓ *PERCENTAGE*: 79.8/100

Matriculation 2017

Agrasen SCH Patel Ngr Bhurkunda Hazaribagh(Jharkhand), India

- ✓ *CGPA*: 8.8/10

RESEARCH & PROJECTS

1. Solving High-Dimensional Differential Equations (Ph.D. Research)

- ✓ Utilizing the Feynman-Kac formula and Backward Stochastic Differential Equations (BSDEs).
- ✓ Applying numerical methodologies to solve the Black-Scholes model.
- ✓ Exploring Physics-Informed Neural Networks (PINNs) for solving high-dimensional heat equations.

2. A Machine Learning and Finite Element Framework for Inverse Elliptic PDEs via Dirichlet-to-Neumann (DtN) Mapping.

3. Comparative Study of Manual Optimizers and Support Vector Machines on the MNIST Dataset.

4. A Compact Fourth-Order Finite Difference Scheme for The Steady Incompressible Navier-Stokes Equations.

5. Higher Order Compact Scheme for Solving Elliptic PDE.

TEACHING EXPERIENCE

Teaching Assistant 2024 – Present

Indian Institute of Technology, Goa, India

- ✓ **MA101:** Calculus
- ✓ **MA203:** Numerical Analysis

PRESENTATIONS

- ✓ Physics-Informed Neural Networks (PINNs) and Feynman-Kac for Solving High-Dimensional Heat Equations, Indian Institute of Technology, Goa, India (Ph.D. Presentation)
- ✓ Comparative Study of Manual Optimizers and Support Vector Machines on the MNIST Dataset, Indian Institute of Technology, Goa, India

AREA OF INTEREST

- ✓ Scientific Machine Learning (SciML) & Physics-Informed Neural Networks (PINNs).
- ✓ Numerical Methods for PDEs (Finite Element & Finite Difference Methods).
- ✓ Inverse Problems in Partial Differential Equations.
- ✓ Stochastic Differential Equations & Mathematical Finance.
- ✓ Computational Fluid Dynamics (CFD).
- ✓ High-Performance Computing (HPC) for Scientific Applications.

SCHOLASTIC ACHIEVEMENTS

- ✓ NET-2024 (PH.D ONLY)
- ✓ GATE-2024 AIR-1020.
- ✓ GATE-2023 AIR-1362.
- ✓ IIT JAM-2022 AIR-528.

SKILLS

Computational

- ✓ MATLAB.
- ✓ Python.
- ✓ Fortran.
- ✓ C++.
- ✓ C.
- ✓ Data Structure in C.
- ✓ FEniCS.
- ✓ High-Performance Computing (HPC) & Parallel Computing (Currently Learning).

Typesetting

- ✓ LaTeX.
- ✓ MS Word.

Languages

- ✓ Hindi.
- ✓ English.

CO-CURRICULAR ACTIVITIES

- ✓ Student Co-Coordinator at Ek Bharat Shreshtha Bharat Campaign, Indian Institute of Technology Bhubaneswar. (2022-2024)
- ✓ Coordinated many departmental cultural festivals during M.Sc. (2022-24).

REFERENCES

Dr. Lok Pati Tripathi
Assistant Professor,
Department of Mathematics and Computer Science,
F-5, Academic Block A (IIT Goa, Goa, India)
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Department of Mathematics and Computer Science,
F-19, Academic Block B (IIT Goa, Goa, India)
Email ID: saumya@iitgoa.ac.in

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Department of Mathematics, School of Basic Sciences (IIT Bhubaneswar, Odisha, India)
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